Qi-Hang Sun | Curriculum Vitae

https://github.com/XZhoou https://xzhoou.top
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RESEARCH INTERESTS

Transportation network analysis (TNA), Travel demand management (TDM), Pedestrian dynamics

EDUCATION

Beihang University **Beijing, CHN**
Ph.D. Candidate in Management Science and Engineering; GPA: 3.9/4.0; Rank: 2/51 *2023.09 - Present*
Doctoral Thesis Topic: *Design and operational optimization of reservation schemes for congestion management*

Beihang University **Beijing, CHN**
B.S. in Biomedical Engineering (Minor) *2020.09 - 2023.06*

Beihang University **Beijing, CHN**
B.S. in Information Management and Information System *2019.09 - 2023.06*

AWARDS & HONORS

- **Joint-training Ph.D. Program Scholarship**, China Scholarship Council, 2025
- **Excellent Student Cadre (Graduate)**, Beihang University, 2025
- **The First Prize Academic Scholarship (Graduate)**, Beihang University, 2024, 2025
- **Merit Student (Graduate)**, Beihang University, 2024
- **Excellent League Member (Graduate)**, Beihang University, 2024, 2025
- **Best Paper Award**, The 5th Frontier Symposium on Traffic Behavior and Transportation Science, 2024
- **Freshmen Scholarship**, Beihang University, 2023
- **Outstanding Undergraduate Graduates**, Beihang University, 2023
- **Merit Student (Undergraduate)**, Beihang University, 2022
- **Model Student of Academic Records**, Beihang University, 2022
- **The Grand Prize Academic Scholarship (Undergraduate)**, Beihang University, 2022
- **The First Prize Discipline Competition Scholarship**, Beihang University, 2022
- **The Second Prize Social Work Scholarship**, Beihang University, 2022
- **Honorable Mention Prize in Mathematical Contest In Modeling (MCM)**, COMAP, 2022
- **The Second Prize in China Undergraduate Mathematical Contest in Modeling**, CUMCM, 2021
- **The Second Prize Academic Scholarship (Undergraduate)**, Beihang University, 2021
- **Outstanding Volunteer**, Beihang University, 2021

INTERNSHIP EXPERIENCE

Ministry of Industry and Information Technology *2022.03-2022.09*
Undergraduate Research Assistant

International Relations Department *2025.09-2026.01*
Student Assistant

SKILLS

Programming	Python, R, Matlab, C, SQL, LaTeX
Languages	Chinese (*Native*), English (*Conversational*)
Software	MS Office, IBM SPSS
Sports	Football, Badminton, Table tennis

PUBLICATIONS

- **New insights into bi-directional pedestrian flows with heterogeneous speeds: modeling and analyses**
  Accepted by The 14th Workshop on Computational Transportation Science (*CTS 2023*)
  Qi-Hang Sun, Ren-Yong Guo
- **Morning Peak Reservation Scheme Incorporating Noncompliance Behaviors: Modeling and Analysis**
  Accepted by The 5th Frontier Symposium on Traffic Behavior and Transportation Science (*TBTS 2024*)
  Qi-Hang Sun, Ren-Yong Guo
- **Flexible reservation scheme for urban transportation network management in a multimodal context**
  Accepted by International Symposium on Multimodal Transportation (*ISMT 2025*)
  Qi-Hang Sun, Ren-Yong Guo
- **Reservation scheme for congestion management during morning commute: Considering noncompliance**
  Submitted to Transportation Research Part C: Emerging Technologies
  Qi-Hang Sun, Ren-Yong Guo
- **Efficiency of Partial Reservation in the Morning Commute under Noncompliance**
  Working paper
  Qi-Hang Sun, Ren-Yong Guo
- **To violate or comply: strategic speculation in reservation scheme**
  Working paper
  Qi-Hang Sun, Ren-Yong Guo

CONFERENCE TALKS

- **The 14th Workshop on Computational Transportation Science**, Shanghai, 2023.08
  Oral Presentation
- **The 5th Frontier Symposium on Traffic Behavior and Transportation Science**, Yichang, 2024.08
  Oral Presentation
- **The 16th Workshop on Computational Transportation Science**, Wuhan, 2025.07
  Oral Presentation
- **The International Symposium on Multimodal Transportation**, Guangzhou, 2025.11
  Oral Presentation

SELECTED COURSES

- **Ph.D. Student:**
  Advanced Microeconomics Theory (94), Advanced Mathematical Programming (93), Optimization Method (98), Advanced Optimization Theory and Algorithm (92), Urban Economics (95), Advanced Academic English Writing (95)
- **Undergraduate:**
  Information Systems Analysis and Design (95), Program Designing (92), Production and Operations Management (94), Artificial Intelligence and Deep Learning (97), Decision-Making and Business Intelligence Systems (96), Large-Scale Database Application Experiment (96)